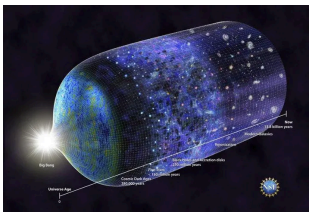
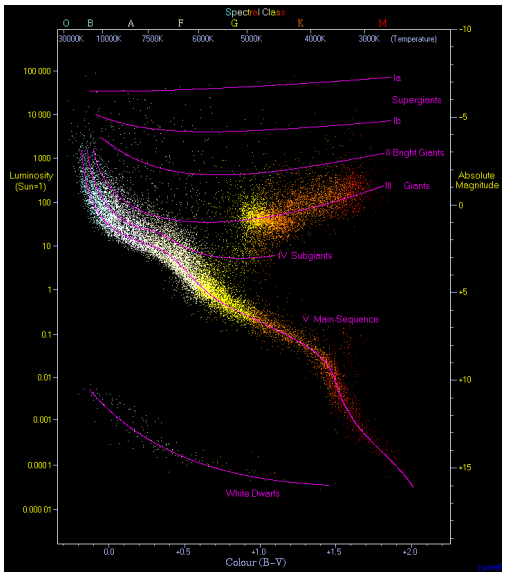
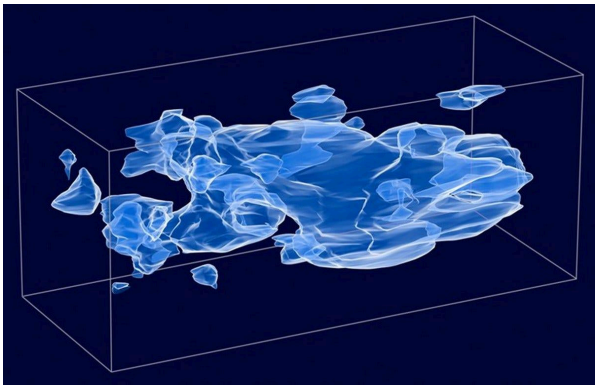
Advanced Astronomy: Universe

- Solar neighborhood and exoplanets
- Stellar formation and evolution
- Star death:
  - Novea
  - Supernovae
  - Neutron stars
  - Black Holes
- Galaxies
  - Dark matter
- Cosmology
  - Big bang
  - Dark energy



In the process, we will learn astrophysics concepts and how to apply them including



- Waves
  - Doppler effect
  - Spectrum
  - Blackbody radiation
- Quantum theory: Bohr model of atomic structure
- Spectroscopy: The interaction of atoms and radiationAstrophysics of multi-body systems
- Special and general relativity
  - Time dilation and length contraction
  - Event horizons
- Nuclear and particle physics
  - (Beyond?) the Standard Model of Particle Physics
  - Nuclear fusion
  - Electron degeneracy and the Chandrasekar mass
  - Neutron degeneracy
- Nuclear and Particle physics II
  - The Standard Model of Particle Physics